COMICCRAFT — SOLUTION REQUIREMENTS

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FUNCTIONAL REQUIREMENTS

FR-1: User can enter a story prompt

FR-2: User can enter main character name

FR-3: User can select a setting (forest, city, space, school, ocean, mountain)

FR-4: User can select a story tone (dramatic, funny, light-hearted, poetic, mysterious)

FR-5: User can select an art style (anime, comic book, realistic, pixel art, watercolor)

FR-6: System generates a 5-panel comic outline using AI

FR-7: System generates narration and dialogue for each panel

FR-8: System generates an illustration for each of the 5 panels

FR-9: System displays all panels with title, image, scene description, and narration

FR-10: User can export the comic as a PDF

NON-FUNCTIONAL REQUIREMENTS

NFR-1: Total generation time under 90 seconds

NFR-2: Works on any modern browser (Chrome, Edge, Firefox)

NFR-3: No user authentication required

NFR-4: API key stored securely in .env

NFR-5: Graceful error handling for API failures and rate limits

NFR-6: Fallback text if an image fails to generate

CONSTRAINTS

- Must use FastAPI for the backend (per TN Skills spec)
- Must use Google Gemini for text generation
- Free image generation (no paid API)
- Runs on Windows with Python 3.10+
- Total cost: Rs. 0